Syllabus: MIST 5775: Cyber Threat Intelligence

Spring 2026

I. Instructor Information

- **Instructor:** Dr. John-David Rusk, CISSP, GCFE
- **Email:** jdrusk@uga.edu
  (Include the course name and your last name in the subject line.)
- **Office Location:** Amos Hall B200G
- **Office Hours:** Monday and Wednesday, from 12:00 pm to 1:00 pm
  (Other times by appointment)

II. MIST 5775: Cyber Threat Intelligence, Course Information

- **Semester and Academic Year:** Spring 2026
- **Effective/Last Revised:** Spring 2026
- **Course Description:**
  Cyber threat intelligence (CTI) aims to provide valuable intelligence to help organizations be aware of and protect against cyber-attacks. This course provides students with a hands-on introduction to the fundamental concepts and tools of cyber threat intelligence.
- **Course Learning Outcomes:**
  - Students will become familiar with the cyber threat intelligence lifecycle, identifying, collecting, and integrating intelligence feeds, common intelligence formats, and standard cyber threat intelligence technologies such as Collective Intelligence Frameworks (CIF's), Trusted Automated eXchange of Intelligence Information (TAXII), and Security Information and Event Management (SIEM). In addition to these traditional topics, this course will deliver cutting-edge cyber threat intelligence education by placing a heavy emphasis on the application and development of state-of-the-art Big Data, visualization, web mining, and machine learning techniques and technologies to bolster common cyber threat intelligence data sources and threat analytics.
  - Students will gain significant experiences in developing useful, actionable, and comprehensive cyber threat intelligence by creating a novel cyber threat intelligence platform utilizing the aforementioned analytics techniques on traditional intelligence feed data and hacker community data sets (e.g., DarkNet marketplaces, Hacker Forums, IRC channels).
  - Upon the successful completion of this course, students will be able to recognize, describe, identify, discuss, explain, and implement cyber threat intelligence

concepts and mechanisms from both technical and managerial perspectives within organizations.

- **Topical Outline**
  - Topic 1: Cyber Threat Intelligence Introduction and Intelligence Strategy
    • Fundamental cyber threat intelligence concepts and vocabulary
    • Intelligence lifecycles
    • Threat trending
    • Critical asset identification
    • Indicators of compromise (IoC's)
    • Modeling threat scenarios for an organization (diamond modelling)
    • Intelligence buy-in
  - Topic 2: Intelligence Collection, Aggregation, and Capabilities
    • Identifying intelligence sources and feeds
    • Collecting and aggregating open-source intelligence using web mining techniques
    • Collecting and aggregating internal intelligence
    • Intelligence formats (e.g., STIX, TAXII, CybOX, MAEC, CAPEC)
    • Setting up collective intelligence framework (CIF) and trusted automated eXchange of intelligence Information (TAXII) servers
    • Security information and event management (SIEM)
    • Big Data technologies (Hadoop and Spark) for CTI data aggregation and processing
  - Topic 3: Threat Analytics and Reporting
    • Lockheed Martin cyber kill chain
    • Internal and external hacker profiling and tracking
    • Fundamental CTI analytics: summary statistics, event correlation, reputation, malware analysis, anomaly detection, forensics
    • Threat analytics utilizing machine learning and text mining approaches
    • Network science applications for cyber threat intelligence applications
    • Threat data visualizations and dashboards
  - Topic 4: Operational Intelligence
    • Actionable intelligence
    • Courses of action
    • Proactive defense
    • Threat dissemination and intelligence sharing
    • Intelligence strategy, process, and systems review

III. Course Policies and Materials

- **Required Course Materials:**
  Course materials will be provided.

- **Grading Policy:**
  Exam 1, 20%
  Exam 2, 20%

Group Presentation, 20%
Weekly Quizzes, 20%
Assignments, 12%
Group Peer Evaluations, 4%
Professional Development, 4%
Total, 100%

- **Attendance Policy:**
  Class sessions will be in-person. Attendance is strongly encouraged but not graded.

- **No Exam Readmittance Policy:**
  To maintain exam integrity, once a student exits the exam room for any reason, reentry is not permitted. Leaving the room constitutes completion and submission of the exam. Exceptions will be considered only in documented emergencies and solely at the instructor's discretion.

IV. Academic Honesty Statement

Academic Honesty / Integrity Expectations:
As a University of Georgia student, you have agreed to abide by the University's academic honesty policy, "A Culture of Honesty," and the Student Honor Code. All academic work must meet the standards described in the "Culture of Honesty" found at http://ovpi.uga.edu/academic-honesty. UGA Student Honor Code: "I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others." A Culture of Honesty, the University's policy and procedures for handling cases of suspected dishonesty, can be found at www.uga.edu/ovpi.

In this course:

1. You are NOT to receive ANY outside assistance on the exams, quizzes, projects, etc., or assignments without prior approval from the professor. Except when noted by the instructor, your assignments must be the result of your individual effort.

2. Unless explicitly stated, artificial intelligence-based technologies, such as ChatGPT, must NOT be used to generate responses for student assignments, exams, quizzes, projects, or any other graded course components.

Students are expected to adhere strictly to the University Honor Code and Academic Honesty Policy and to follow generally accepted standards of honesty in all assignments and examinations. Violations of academic integrity may result in a grade of zero (0) for the assignment and/or a final grade of "F" in the course.

Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation. Questions related to course assignments and the academic honesty policy should be directed to the instructor.

Use of AI for Coursework:

Generative artificial intelligence (GAI) and AI editing software are powerful tools that can enhance learning and communication when used appropriately. In this course, you are welcome to explore the use of GAI tools in a limited capacity. AI-based editors, such as those in Microsoft Word and Grammarly, may be used to edit drafts of your work. GAI may be used, but only in the capacity of brainstorming ideas or organizing your thoughts. Still, use of GAI tools should be limited to providing support as you develop your thinking and knowledge base for assignments. GAI should never be used to write entire drafts or sections of any assignment.

Please note that you may not represent output generated by a GAI tool as your own work. Any such use of GAI output must be appropriately cited or disclosed, including quotation marks and in-line citations for direct quotes. Use of AI outside of the allowed uses mentioned above will be considered unauthorized assistance and a violation of the Academic Honesty Policy. Suspected unauthorized assistance, plagiarism, or other violations of UGA's "A Culture of Honesty" will be reported to the Office of Academic Honesty. For full details on how to properly cite AI-generated work, please see the APA Style article, How to Cite ChatGPT.

At times, you may be unsure where the line is between collaborating with GAI and copying from GAI. In that case, I recommend that you do not have your assignment and the GAI tool open on your device at the same time. Instead, take notes in your own words while you interact with the GAI tool, then use your notes to remind you of what you've learned and to inform your work. Never copy output from GAI tools into your assignment. Instead, use your interaction with the tool as a learning experience, then close the interaction down, open your assignment, and let your assignment reflect your improved understanding.

Finally, GAI is highly vulnerable to inaccuracy and bias. You should assume GAI output is wrong unless you either know the answer or can verify it with another source. It is your responsibility to assess the validity and applicability of any GAI output used.

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A Culture of Honesty:
A Culture of Honesty, the University's policy and procedures for handling cases of suspected dishonesty, can be found at www.uga.edu/ovpi.

UGA Well-being and Student Support Resources

UGA Well-being Resources promote student success by cultivating a culture that supports a more active, healthy, and engaged student community.

Anyone needing assistance is encouraged to contact **Student Care & Outreach (SCO)** at 706-542-8479 or sco.uga.edu. SCO helps students navigate difficult circumstances by connecting them with appropriate resources or services.

UGA provides both clinical and non-clinical options to support student well-being and mental health, any time, any place.

Resources:

- Well-being Resources
- Student Care and Outreach
- University Health Center
- Counseling and Psychiatric Services (CAPS) — 24/7 crisis support: 706-542-2273
- Health Promotion / Fontaine Center
- Disability Resource Center & Testing Services

Additional information, including free digital well-being resources, is available through the UGA app or well-being.uga.edu.

Statement on Inclusive Excellence

The Terry College of Business is committed to promoting an inclusive learning and working environment among its students, faculty, and staff. This class welcomes the open exchange of ideas and provides a professional environment that recognizes the inherent worth of every person. It aims to foster dignity, understanding, and mutual respect among all individuals in the class.

The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.